Assignment 7: SQL Joins & Set Operations

Overview:

You will define two tables—Employees and Departments—and practice a variety of SQL joins and set operations to combine and compare their data.

1. Table Creation & Data Population

```sql
-- Employees table
CREATE TABLE Employees (
  EmpID INT PRIMARY KEY,
  EmpName VARCHAR(50),
  DepartmentID INT
);
INSERT INTO Employees (EmpID, EmpName, DepartmentID) VALUES
  (1, 'Alice',   101),
  (2, 'Bob',     102),
  (3, 'Charlie', 103),
  (4, 'Diana',   NULL),
  (5, 'Eve',     101);
```

Submission Type : Write the SQL query in the **PDF/Doc/Word** file and Take a screenshot of this part Output **(as screenshot_1a)** and save it in the same **PDF/Doc/Word** file.

```sql
-- Departments table
CREATE TABLE Departments (
  DepartmentID INT PRIMARY KEY,
  DeptName VARCHAR(50),
  Location VARCHAR(50)
);
INSERT INTO Departments (DepartmentID, DeptName, Location) VALUES
  (101, 'HR',    'New York'),
  (102, 'IT',    'San Francisco'),
  (103, 'Finance',  'Chicago'),
  (104, 'Sales',          'Boston');
```

Submission Type : Write the SQL query in the **PDF/Doc/Word** file and Take a screenshot of this part Output **(as screenshot_1b)** and save it in the same **PDF/Doc/Word** file.

2. SQL Tasks

1. **a) Inner Join:** List all employees with assigned department names.

SQL:

```
SELECT E.EmpID, E.EmpName, D.DeptName
FROM Employees E
INNER JOIN Departments D
  ON E.DepartmentID = D.DepartmentID;
```

Submission Type : *Write the SQL query in the* ***PDF/Doc/Word*** *file and Take a screenshot of this part Output* ***(as screenshot_2a)*** *and save it in the same* ***PDF/Doc/Word*** *file.*

2. **b) Left Join:** Show all employees, with department name if assigned (NULL if not).

SQL:

```
SELECT E.EmpID, E.EmpName, D.DeptName
FROM Employees E
LEFT JOIN Departments D
  ON E.DepartmentID = D.DepartmentID;
```

Submission Type : *Write the SQL query in the* ***PDF/Doc/Word*** *file and Take a screenshot of this part Output* ***(as screenshot_2b)*** *and save it in the same* ***PDF/Doc/Word*** *file.*

3. **c) Right Join:** Show all departments, with employee name if exists (NULL if none).

SQL:

```
SELECT D.DepartmentID, D.DeptName, E.EmpName
FROM Employees E
RIGHT JOIN Departments D
  ON E.DepartmentID = D.DepartmentID;
```

Submission Type : *Write the SQL query in the* ***PDF/Doc/Word*** *file and Take a screenshot of this part Output* ***(as screenshot_2c)*** *and save it in the same* ***PDF/Doc/Word*** *file.*

4. **d) Full Outer Join:** Include all employees and departments, matching where possible.

SQL:

```
SELECT E.EmpID, E.EmpName, D.DeptName
FROM Employees E
FULL OUTER JOIN Departments D
  ON E.DepartmentID = D.DepartmentID;
```

Submission Type : *Write the SQL query in the* ***PDF/Doc/Word*** *file and Take a screenshot of this part Output* ***(as screenshot_2d)*** *and save it in the same* ***PDF/Doc/Word*** *file.*

5. **e) Cross Join:** Generate every combination of employee names and department names.

SQL:

```
SELECT E.EmpName, D.DeptName
FROM Employees E
CROSS JOIN Departments D;
```

Submission Type : *Write the SQL query in the* ***PDF/Doc/Word*** *file and Take a screenshot of this part Output* ***(as screenshot_2e)*** *and save it in the same* ***PDF/Doc/Word*** *file.*

6. **f) Union All:** Create a list combining all employee names and department names in one column.

SQL:

```
SELECT EmpName AS Name FROM Employees
UNION ALL
SELECT DeptName FROM Departments;
```

Submission Type : *Write the SQL query in the* ***PDF/Doc/Word*** *file and Take a screenshot of this part Output* ***(as screenshot_2f)*** *and save it in the same* ***PDF/Doc/Word*** *file.*

7. **g) Intersect:** Find department IDs present in both tables.

SQL:

```
SELECT DepartmentID FROM Employees
INTERSECT
SELECT DepartmentID FROM Departments;
```

Submission Type : *Write the SQL query in the* ***PDF/Doc/Word*** *file and Take a screenshot of this part Output* ***(as screenshot_2g)*** *and save it in the same* ***PDF/Doc/Word*** *file.*

8. **h) Except:** List department IDs in Departments but not in Employees.

SQL:

```
SELECT DepartmentID FROM Departments
EXCEPT
SELECT DepartmentID FROM Employees;
```

Submission Type : *Write the SQL query in the* ***PDF/Doc/Word*** *file and Take a screenshot of this part Output* ***(as screenshot_2h)*** *and save it in the same* ***PDF/Doc/Word*** *file.*

> Tip: If you encounter a "table already exists" error, append a suffix to the table name (e.g., Employees1).

Submission Guidelines:

Create One Main Folder (named Assignment7_YourName) which includes -

- One **Doc/PDF/Word** file containing all the Queries and the screenshots of each part.
- Put your SQL file as **Assignment7_YourName.sql**

Convert the folder into ZIP format.
Upload the ZIP folder on the submission portal.

Watch this to learn how to create ZIP file

https://youtu.be/c7x-Q3AQEgw?si=_mS4c7OXADQ7DroE

If you have any doubt, you can directly contact the Tutedude mentor.